

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Regular/Supplementary Winter Examination – 2024**

Course: B.Tech

Branch: Mechanical Engineering/Mechanical Engineering Sandwich**Semester: V****Subject Code & Name: BTMC506 Applied Thermodynamics****Max Marks: 60****Date: 25/02/2025****Duration: 3 Hr.****Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/ CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	The mass of oxygen required to convert 1 kg of carbon to CO ₂ will be (a) 3/11 kg (b) 11/3 kg (c) 3/8 kg (d) 8/3 kg	CO1	1
2	Orsat apparatus is used to determine in flue gases the percentage of (a) CO ₂ , CO, and O ₂ (b) CO ₂ , CO, O ₂ and N ₂ (c) CO ₂ (d) CO	CO2	1
3	Consider the following: 1. Safety valve 2. Steam trap 3. Steam separator 4. Economizer. Among these, the boiler accessories would include. (a) 1, 2, and 3 (b) 2, 3, and 4 (c) 1 and 4 (d) 1, 2, 3, and 4	CO3	1
4	The draught produced by the chimney is called (a) Natural (b) Induced (c) Forced (d) Balanced	CO3	1
5	The draught in locomotive is produced by (a) Forced fan (b) Induced fan (c) Steam jet (d) Chimney	CO2	1
6	Consider the following statements: The reheat cycle helps to reduce I. fuel consumption, II. steam flow, III. the condenser size Which of these statements are correct? (a) I and II (b) I and III (c) II and III (d) I, II, and III	CO2	1
7	The function of the surface condenser is to (a) lower the engine thermal efficiency (b) Increase the engine thermal efficiency (c) Increase the back pressure of the engine (d) cool the exhaust gases	CO3	1
8	A single-stage impulse turbine with a diameter of 120 cm runs at 300 rpm. If the blade speed ratio is 0.42, then, the inlet velocity of steam will be	CO4	1

	(a) 79 m/s	(b) 188 m/s	(c) 450 m/s	(d) 900 m/s		
9	What is the cause of reheat factor in a steam turbine?				CO3	1
	(a) Reheating	(b) Superheating	(c) Supersaturation	(d) Blade friction		
10	In a simple impulse turbine, the nozzle angle at the entrance is 30° . What will be the blade-speed ratio for maximum diagram efficiency?				CO4	1
	(a) 0.433	(b) 0.25	(c) 0.5	(d) 0.75		
11	Which one of the following statements is correct? In reciprocating compressors, one should aim at compressing the air				CO3	1
	(a) adiabatically	(b) isentropically	(c) isothermally	(d) polytropically		
12	The p-V diagram shown in below Figure is for				CO3	1
	(a) Roots blower	(b) Vanes blower	(c) Centrifugal compressor	(d) axial compressor		
Q. 2	Solve the following.					12
A)	Develop the equation for the measurement of calorific value of solid fuel by the use of bomb calorimeter with neat sketch.				CO2	6
B)	How do you classify the fuels? Develop the combustion equation of hydrogen on mass basis.				CO2	6
Q.3	Solve the following.					12
A)	A boiler produces 4000 kg of dry and saturated steam in 6 hours. In the same time the mass of coal consumed is 300 kg of coal. The mean pressure of steam generated is 11 bar. The temperature of the feed water is 35°C . Determine the boiler efficiency if the calorific value of the fuel is 30000 kJ/kg.				CO3	6
B)	Classify the boiler mountings with their functions.				CO3	6
Q. 4	Solve Any Two of the following.					12
A)	Derive the equation for thermal efficiency of Joule cycle with neat sketch.				CO2	6
B)	Explain the supersaturated flow through the nozzle with schematic diagram.				CO2	6
C)	A reheat cycle operating between 30 bar and 0.04 bar has a superheat and reheat temperature of 450°C . The first expansion takes place till the steam is dry saturated and then reheat is given. Determine the ideal cycle efficiency neglecting feed pump work.				CO3	6

Q.5	Solve Any Two of the following.		12
A)	Explain the reheat factor for steam turbine with suitable schematic diagram.	CO4	6
B)	Explain the different elements of condensing plant with suitable schematic diagram.	CO4	6
C)	Steam at 4.9 bar and 160°C is supplied to a single stage impulse turbine at the rate of 60 kg/min. From there, it is exhausted to a condenser at a pressure of 0.196 bar. The blade speed is 300 m/s. The nozzles are inclined at 25° to the plane of the wheel and outlet blade angle is 35°. Neglect friction losses and estimate (a) the theoretical power developed by the turbine, (b) the diagram efficiency, and (c) the stage efficiency.	CO4	6
Q. 6	Solve Any Two of the following.		12
A)	Explain reciprocating air compressor with neat sketch.	CO5	6
B)	Explain the benefits of the multi staging in reciprocating air compressor with PV diagram.	CO5	6
C)	Derive the equation for indicated work of single stage, single acting reciprocating air compressor with neat sketch.	CO5	6
	*** End ***		